$$A^0, \epsilon \in \mathbb{R}^{H \times d}, \quad \epsilon \sim \mathcal{N}(0, I)$$

$$\overline{\alpha}_k = \prod_{j=1}^{k} (1 - \beta_j)$$

$$A^k = \sqrt{\overline{\alpha}_k}\, A^0 + \sqrt{1 - \overline{\alpha}_k}\, \epsilon$$

$$c = \text{Encoder}(O)$$

$$\mathcal{L} = \mathbb{E}_{A^0, O, k, \epsilon} \left[\frac{\| \epsilon - \epsilon_\theta(A^k, k, c) \|_F^2}{Hd} \right]$$